

FarmBook User Stories

Product backlog with priorities, estimates, acceptance criteria and sprint plan (updated week 9)

How the backlog is prioritised and estimated

Priority (MoSCoW):

Must: needed for FarmBook to be usable at all and other stories depend on it (accounts, core records, saving data).

Should: important and expected, but the app still works without it.

Could: valuable extras that are scheduled only once Must and Should stories be done.

Wont: agreed to leave for a future release.

Estimate (story points): Relative effort on the Fibonacci scale (1, 2, 3, 5, 8). Login for example is a 3-point reference story, one screen, one table and one query. A new record type that follows the existing DAO + table + form pattern is 3, stories that add a new table linked to existing records are 5, summary and trend views that combine several data sets are 8.

Acceptance criteria: Each story is done only when all of its criteria pass, the code is merged through a pull request and related unit tests pass.

Status: Taken from the code in the repository at Week 9.

Done: all criteria met.

In progress: part of the story is built.

Not started: no code yet.

User stories

User stories for: FR01 - User Access

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Log In to FarmBook	As a farmer, I want to be able to login with my credentials so that my farm records are not visible to anyone else who uses the same computer.	1	With a registered account, when I enter the correct username and password I am then taken to the dashboard.	MUST	3 story points	Let's a registered farmer open their own farm records securely. The entered username and password are hashed and checked against the users table and a session is started. Sprint: Prototype (W1-9) Status: Done
		2	Given a wrong username or password when I log in, I am then given an error and I stay on the login page.			
		3	Pages other than login/register cannot be opened without logging in.			
Log Out	As a farmer, I want to be able to logout so that I can leave my computer without exposing my data.	1	When I click Logout, my session ends and I am returned to the login page.	MUST	1 story point	Ends the session so no farm records are left visible on a shared computer. Every page checks the session, so logging out returns the farmer to the login screen. Sprint: Prototype (W1-9) Status: Done
		2	After logging out, I must log in again to see any farm records.			
Register an Account	As a farmer, I want to be able to register an account with a unique username and use my email address for recovery.	1	A username or email that is already registered is rejected with a message.	MUST	3 story points	The onboarding step for new farmers. Creates an account with a unique username, a valid
		2	An invalid email format is rejected.			
		3	The password is stored hashed and never as plain text.			

		4	Email recovery: a password reset can be requested using the registered email.			email and a hashed password and keeps the email for password recovery later. Sprint: Prototype (W1-9), recovery in W12-13 Status: In progress
Farm Dashboard	As a farmer, I want to be able to have a dashboard tied to my account that shows all the details of my farm like crops, inventory, stock movement and information related to my animals.	1	After login, the dashboard shows the logged-in username.	SHOULD	5 story points	The home screen after login. It is the central place to reach every feature and gives a quick snapshot of the farm. Sprint: Prototype (W1-9), data in W10-11 Status: In progress
		2	The dashboard links to Crops, Inventory and Livestock and Settings.			
		3	The dashboard shows a summary of my crops, stock and animals.			
Manage Account Settings	As a farmer, I want to be able to manage my account information and dashboard settings.	1	From Account/Settings I can change my email and password.	COULD	5 story points	Let's the farmer keep their details up to date and change their password. Lower priority because the app is fully usable without it. Sprint: W13 (stretch) Status: Not started
		2	A password change requires my current password.			
		3	Changes are saved and still apply after logging out and back in.			

User stories for: FR02 - Crop Records

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Record Planted Crops	As a farmer, I want to be able to record crops that I've planted that include, type, field or plot, quantity and the date it was planted.	1	I can add a crop with type, field/plot, quantity and date planted.	MUST	3 story points	The core crop records every other crop feature builds on. Captures what was planted, where, how much and when. Sprint: Prototype (W1-9), field/plot in W10-11 Status: In progress
		2	The new crop appears in the crop table straight away.			
		3	A crop cannot be saved with a missing field or a non-numeric quantity.			
Add Crop Notes	As a farmer, I want to be able to write notes about my crops to detail any special cases.	1	I can add and edit an optional notes field on a crop.	SHOULD	1 story point	A free-text field for anything unusual about a crop, such as disease, weather damage or special treatment. Sprint: Prototype (W1-9) Status: Done
		2	Notes are shown in the crop table and saved with the crop.			

User stories for: FR03 - Crop Progress

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Log Crop Growth	As a farmer, I want to be able to log the growth of my crops so I can see the progress over the season.	1	I can add a dated growth entry (e.g. stage, height, notes) to a crop.	SHOULD	5 story points	Builds a timeline of growth entries for each crop so progress can be compared across the season. Needs a new table linked to crops. Sprint: W10-11 Status: Not started
		2	A crop's growth entries are listed in date order.			
Mark Crops Harvested or Failed	As a farmer, I want to be able to mark a crop as harvested or failed so finished crops don't clutter my dashboard.	1	I can set a crop's status to Growing, Harvested or Failed.	SHOULD	2 story points	Adds a status to each crop so finished crops drop out of the active list without losing their history. Sprint: W10-11 Status: Not started
		2	Harvested and failed crops are hidden from the active list but can still be viewed.			

User stories for: FR04 - Crop Needs

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Record Crop Needs	As a farmer, I want to be able to record crop needs such as watering, fertilizer and pest treatment so I don't have to rely on remembering what each crop requires.	1	I can add a need (watering, fertiliser or pest treatment) to a crop with a frequency or due date.	COULD	5 story points	Stores what each crop needs and when, so the farmer does not have to remember watering, fertilising and pest treatment schedules. Sprint: W12-13 Status: Not started
		2	A crop's needs are shown when the crop is selected.			

User stories for: FR05 - Inventory Records

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Add Inventory Items	As a farmer, I want to be able to add seeds and equipment to my inventory with the name, category, unit and quantity so that I know what I have on hand.	1	I can add an item with name, category, unit and quantity.	MUST	3 story points	The base inventory record for seeds and equipment, shown in a sortable table. Stock movement and feed tracking build on it. Sprint: Prototype (W1-9), validation in W10-11 Status: In progress
		2	Items are listed in a table sorted by name.			
		3	A non-numeric quantity is rejected with a message instead of crashing.			

User stories for: FR06 - Stock Movement

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Record Incoming Stock	As a farmer, I want to be able to record incoming stock with quantity, date and optional sources so that my inventory reflects new purchases or deliveries.	1	I can select an item and record a quantity, date and optional source.	MUST	3 story points	Updates stock levels when purchases or deliveries arrive, with an optional source for traceability. Sprint: Prototype (W1-9) Status: Done
		2	The item's quantity increases by the amount received.			
Record Stock Used or Sold	As a farmer, I want to be able to record stock used or sold so that my inventory levels stay accurate while I complete my work.	1	I can select an item and record a quantity used or sold.	MUST	3 story points	Keeps stock levels accurate as items are used or sold and stops quantities going below zero. Sprint: Prototype (W1-9) Status: Done
		2	The item's quantity decreases by that amount.			
		3	Removing more than is in stock is rejected, so quantities never go below zero.			
Stock Overview	As a farmer, I want to be able to have a single view that shows all my stock and their current quantities so that I can check what needs reordering at a glance.	1	One table shows every item with its current quantity, including incoming and outgoing movements.	SHOULD	3 story points	One view of every item and its current level so reordering is easy to spot. Incoming and outgoing stock currently use a separate table from the inventory page, so the two need to be combined. Sprint: W10-11 Status: In progress
		2	Items at or below a low-stock level are highlighted.			

User stories for: FR07 - Livestock Records

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Record Livestock	As a farmer, I want to be able to record animals individually or as a group with species, identifiers and date acquired or born.	1	I can add an animal with species, identifier/tag and date acquired or born.	MUST	3 story points	The base animal record that feed and output tracking link to. Input is checked by LivestockValidator. Sprint: Prototype (W1-9), groups in W12-13 Status: In progress
		2	Missing fields or an invalid date (not YYYY-MM-DD) are rejected.			
		3	Saved animals are listed on the Livestock page.			
		4	I can record a group of animals with a head count.			

User stories for: FR08 - Feed Information

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Record Feeding	As a farmer, I want to be able to record what feed was given to which animals and when so that I can track feed use against my animals.	1	I can record a feed type, quantity, date and the animal or group it was given to.	COULD	5 story points	Links feed given to the animals that received it and reduces feed stock. Sprint: W12-13 Status: Not started
		2	Recording feed reduces that feed item's stock in inventory.			

User stories for: FR09 - Livestock Outputs

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Record Livestock Outputs	As a farmer, I want to be able to record outputs like milk, wool and eggs with quantity and date so I can track productivity and quality over time.	1	I can record an output type, quantity, unit and date against an animal or group.	COULD	3 story points	Tracks what animals produce over time, such as milk, wool and eggs, to show productivity. Sprint: W12-13 Status: Not started
		2	Outputs can be listed by animal or group and date.			

User stories for: FR10 - Resource Consumption

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Record Water Use	As a farmer, I want to be able to record water consumption for a period with the area or activity it relates to so that I can see where my water is going.	1	I can record litres used, a start and end date and the area or activity.	COULD	3 story points	Logs water use by period and by area or activity so the farmer can see where water goes. Feeds the resource summary. Sprint: W12-13 Status: Not started
		2	Water records are listed and can be filtered by area or activity.			
Record Energy Use	As a farmer, I want to be able to record my energy consumption like how many kWh, time frame and type of power it came from (grid, solar, hydro, etc).	1	I can record kWh, a start and end date and the source (grid, solar, hydro, other).	COULD	3 story points	Logs energy use in kWh by period and by source (grid, solar, hydro). Feeds the resource summary. Sprint: W12-13 Status: Not started
		2	Energy records are listed and can be filtered by source.			

User stories for: FR11 - Waste Records

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Record Waste	As a farmer, I want to be able to record waste produced and how it was handled like composted, recycled or disposed of so I can track how much is diverted rather than dumped.	1	I can record a waste type, amount, date and how it was handled.	COULD	3 story points	Logs waste and how it was handled so the share diverted from landfill can be tracked. Sprint: W12-13 Status: Not started
		2	The percentage of waste diverted (composted or recycled) is shown.			

User stories for: FR12 - Resource Summary

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Resource Summary	As a farmer, I want to be able to see all my resource use in a quick easy to view place to see how well I'm tracking.	1	One page shows total water, energy and waste for a chosen period.	COULD	8 story points	Combines water, energy and waste records into one overview. Estimated high because it draws on three data sets. Sprint: W13 (stretch) Status: Not started
Resource Trends	As a farmer, I want to be able to see trends and high usage of water and electricity without looking through every log.	1	A chart shows water and energy use over time.	COULD	8 story points	Charts resource use over time and flags high usage. Deferred because trends need several weeks of recorded data first. Sprint: Future release Status: Not started
		2	Periods of unusually high usage are highlighted.			
Crop Chart	As a farmer, I want to be able to see a chart of my crops to see what I've planted at a quick glance.	1	A button switches the crop table to a pie chart of crops by amount.	SHOULD	3 story points	A pie chart view of crops by amount, switched on from the crops page, for a quick picture of what is planted. Sprint: Prototype (W1-9) Status: Done
		2	The chart updates when crops are added, edited or deleted.			

User stories for: FR13 - Persistent Storage

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Persistent and Secure Storage	As a farmer, I want to be able to have my information retained securely and when I close the program to still be there when I reopen and log back in.	1	Records are saved to a local SQLite database and are still there after restarting the app.	MUST	5 story points	Saves all records in a local SQLite database so they are still there after restarting, with passwords stored as hashes. Records still need to be linked to each user. Sprint: Prototype (W1-9), per-user data in W10-11 Status: In progress
		2	Passwords are stored as SHA-256 hashes.			
		3	Each user only sees their own farm records.			

User stories for: FR14 - Retrieve and Update Records

TITLE	USER STORY	ACCEPTANCE CRITERIA		PRIORITY	ESTIMATION	DESCRIPTION
Find, Correct and Delete Records	As a farmer, I want to be able to find, correct and delete any earlier	1	I can select a crop or inventory item, edit it and save the change.	MUST	5 story points	Let's the farmer fix mistakes so they do not distort records. Crops and inventory can already
		2	I can delete a crop or inventory item.			
		3	I can edit and delete livestock records.			

	entry mistakes so that it does not permanently distort my records.	4	I can search or filter records to find an earlier entry.			be edited and deleted, livestock and searching are still to do. Sprint: Prototype (W1-9), livestock & search in W10-11 Status: In progress
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Summary

Priority	Stories	Story points
Must	10	32
Should	6	19
Could	9	43
Wont	0	0

Status (Week 9)	Stories	Story points
Done	6	14
In progress	8	30
Not started	11	50

Release and sprint plan

Two releases: The prototype (week 9) and the final release (week 13). The remaining weeks are split into two sprints planned at about 30 points each. Stories already in progress are counted at the full estimate so the W10-11 sprint has some slack for the partly finished work remaining.

Release / sprint	Stories	Justification
Prototype (W1-9)	Log In to FarmBook, Log Out, Register an Account, Farm Dashboard (layout), Record Planted Crops, Add Crop Notes, Add Inventory Items, Record Incoming Stock, Record Stock Used or Sold, Record Livestock, Crop Chart, Persistent and Secure Storage, Find, Correct and Delete Records (crops and inventory)	Built the core of every Must-have so the main loop (log in -> record crops, stock and animals -> data saved) works end to end. The parts still missing are marked in progress.
Sprint W10-11	Finish: Record Planted Crops (field/plot and input validation), Add Inventory Items (input validation), Find, Correct and Delete Records (livestock and search), Persistent and Secure Storage (per-user data), Farm Dashboard (farm data), Stock Overview. Start: Log Crop Growth, Mark Crops Harvested or Failed	Close the gaps in Must-haves first, because later features (feed, outputs, summaries) build on crops, stock and livestock records being complete and editable.

Sprint W12-13	Register an Account (simulated email recovery), Record Crop Needs, Record Livestock (groups), Record Feeding, Record Livestock Outputs, Record Water Use, Record Energy Use, Record Waste. Testing, bug fixes and documentation. Stretch goals: Manage Account Settings. Resource Summary.	Should/Could features that each add one record type using the same DAO + table pattern already built, so they are lower risk and can be split across the team. Final bug checking and polishing at the end of the week 13.
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Revision history

Version	Change
v1	Initial backlog: 14 functional requirements (FR01-FR14) with user stories.
v2	Compared backlog against the prototype. Added stories for the dashboard, crop notes and crop chart, which had been built but were not in the backlog.
v3	Added account settings story, extended FR14 to include deleting records, fixed wording (unique username, kWh, FR03/FR13 phrasing).
v4 (W9)	Split requirements into numbered stories, added MoSCoW priorities, story point estimates, acceptance criteria, sprint allocation and current status based on the code in the repository. Planned remaining sprints for weeks 10-13.